

Technical Reference: Papyrus AI Architecture & Line Logic (NR2003 / Retro Sims)

1. Core Architecture: LP (Line Position) Files

Papyrus sims (*Indianapolis 500*, *IndyCar Racing*, *NASCAR Racing* series through *NR2003*) control AI behavior via pre-computed positional line files (.lp files). Rather than relying on real-time neural networks or dynamic pathfinding, the AI executes deterministic spatial logic relative to track boundaries.

- **Coordinate System:** Track data is plotted as relative offsets (lateral displacement) from a primary reference axis—typically the centerline of the racing surface.
- **Closed-Loop Splines:** Lines are defined by sequential coordinate points along the track length. If the start-finish coordinates do not align seamlessly with the end-of-lap coordinates, cars exhibit a visible lateral "snap" or jump when crossing the blend point.

2. Standard LP Line Definitions

Line Type	Designation	Function & Logic
Race Line	Ideal / Fast Line	The baseline single-car trajectory optimizing entry, apex, and track-out speed. AI defaults to this line when in clean air.
Inner Race Line	Min Racing Boundary	Leftward limit of normal side-by-side competition. Dictates car placement when defending or running side-by-side.
Outer Race Line	Max Racing Boundary	Rightward limit of normal competition. Prevents AI from tracking too close to the outer retaining wall.
Inner Panic Line	Emergency Lower Limit	Maximum evasive boundary below the track (often extending onto the apron or paved runoff).
Outer Panic Line	Emergency Upper Limit	Maximum evasive boundary towards grass runoffs, walls, or outer limits.

Pit Line(s)	Pit Entry / Box / Exit	Dedicated path governing deceleration, pit-box transit, acceleration lanes, and rejoin points.
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3. Spatial Awareness & Collision Avoidance

- **Corridor Constraints:** AI vehicles determine validity based on bounding zones between the inner and outer lines:
 - **Racing Zone (Inner to Outer Race Line):** The active track envelope. When cars are side-by-side, lateral offsets are clamped within this band.
 - **Panic Zone (Inner to Outer Panic Line):** Activated when a vehicle or blockage enters the AI lookahead window. AI yields its standard racing line and uses the expanded runoff/apron envelope while shedding speed.
- **Proximity Rules:**
 - If an inside vehicle is detected, the AI holds the outer bounds of the active corridor.
 - If flanked on both sides, the AI maintains a straight median path until boundary contraction forces one vehicle to back out of the throttle.

4. Modding Trade-offs & Emergent Bottlenecks

The "Artificially Induced Three-Wide" Problem

- Modders frequently expanded the distance between [race.lp](#), [minrace.lp](#), and [maxrace.lp](#) across banking transitions to force 3-wide pack racing.
- **Failure State:** Because track geometry and legacy friction logic heavily prioritize the bottom groove, cars stranded in the artificially widened high groove bled longitudinal momentum, triggering continuous shuffling and unnatural pack rotation.

Entry Funneling on Road Courses / Sharp Corners

- Widening the race corridor aggressively on straightaways ahead of a tight corner causes the field to fan out 3-to-4 wide on corner approach.
- **The Accordion Jam:** As lines funnel down into a single apex slot, spatial prioritization forces the field to brake simultaneously to yield right-of-way. This creates a moving bottleneck where human players can exploit the inside lane and pass multiple AI cars regardless of difficulty percentage.

5. Line Creation & Editing Toolflow

- **Tools:** Dedicated utilities such as [LPEdit](#) (DOS-based editor developed by early modders/developers like Nim Cross).
- **Workflow:**
 1. Drive a clean lap in-game following the target line (ideal, pit entry, or extreme boundary).
 2. Save and export the replay telemetry.
 3. Convert replay telemetry data into positional points using extraction tools.
 4. Import coordinates into [LPEdit](#) to inspect the 2D offset graph, smooth inflection points, and blend the lap start/finish coordinates.
 5. Package compiled [.lp](#) files into the circuit directory.

